

STRENGTH TRAINING INFOGRAPHICS COLLECTION

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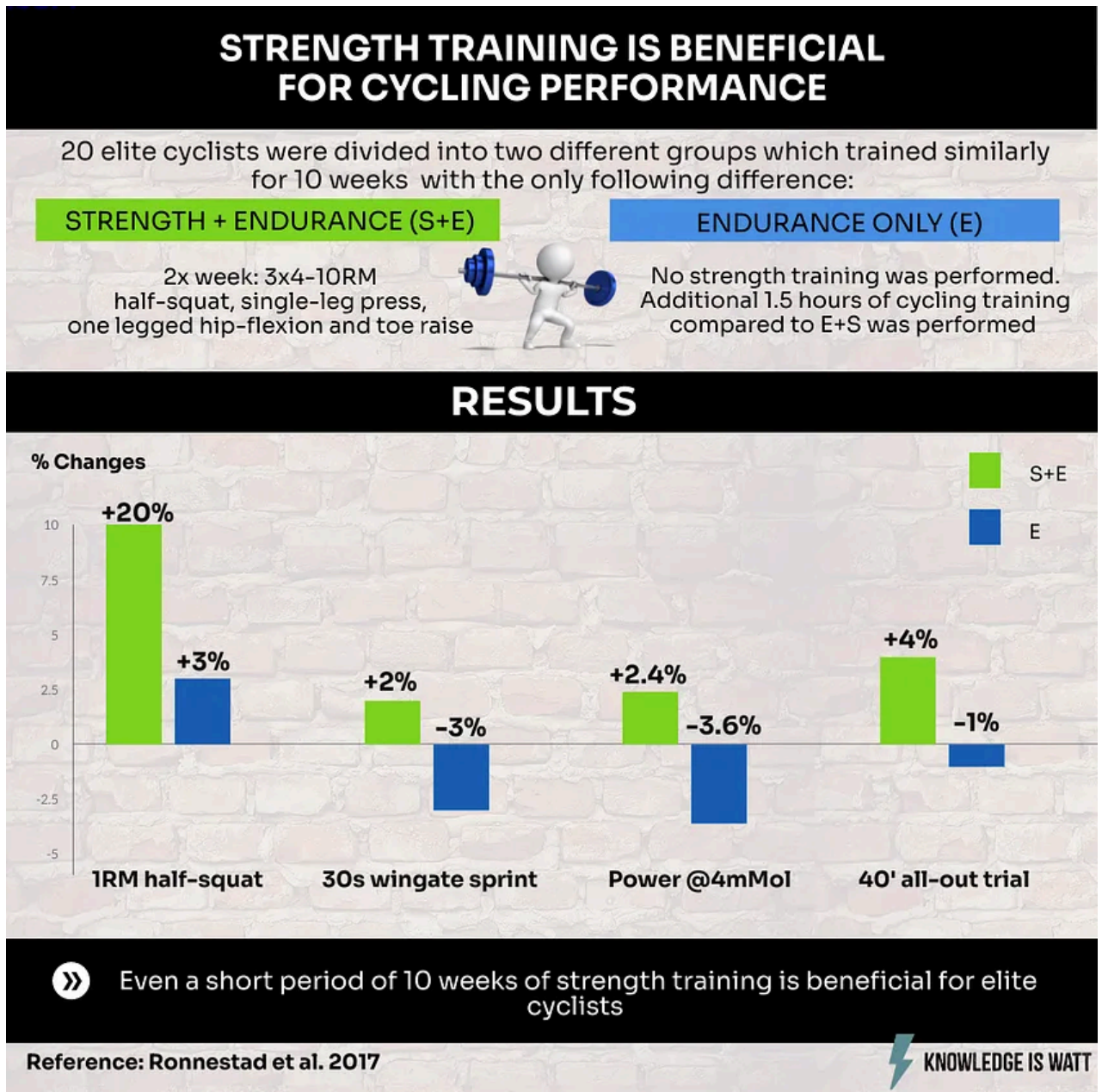
Knowledgeiswatt

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Here is the growing infographics library about all the relevant researches on **strength training and cycling performance** with links to each study's summary.

1. 10 WEEKS OF HEAVY GYM STRENGTH TRAINING DURING PREPARATORY PERIOD IMPROVE CYCLING PERFORMANCE

[Research Summary and Comment](#)



2. HEAVY GYM STRENGTH TRAINING BENEFITS EVEN DURABILITY

[Research Summary and Comment](#)

STRENGTH TRAINING IMPROVES DURABILITY IN WELL-TRAINED CYCLISTS

23 well-trained cyclists (VO₂max ~70) were divided into two different groups which trained similarly on the bike for 12 weeks (low-intensity predominantly) with the only following difference:

STRENGTH + ENDURANCE (S+E)

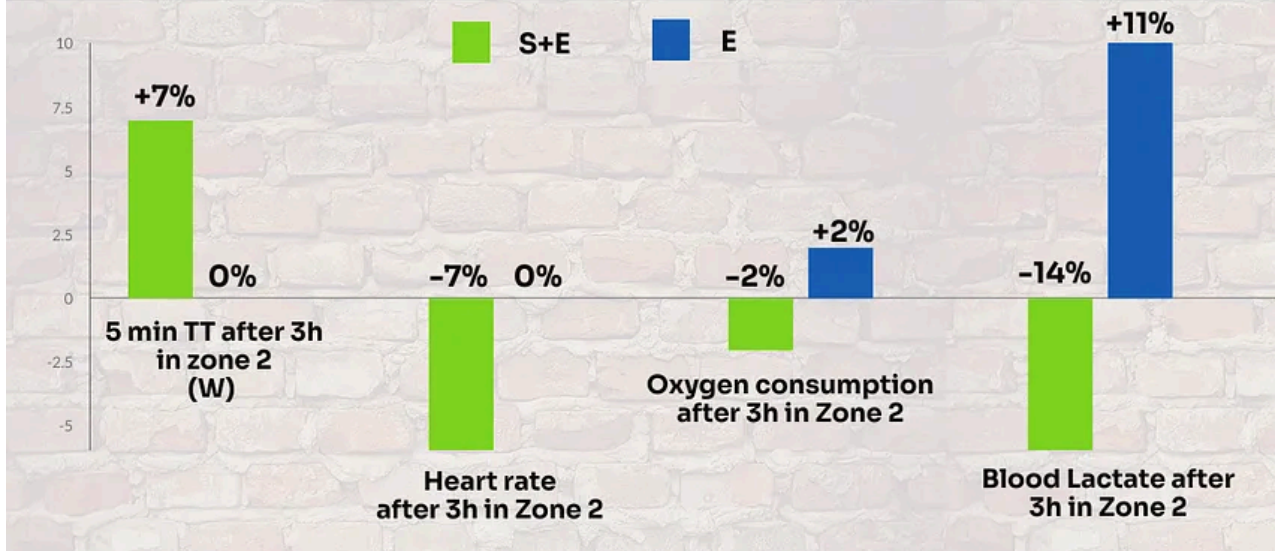
2x week: 3x4-10RM
half-squat, single-leg press,
one legged hip-flexion and toe raise



ENDURANCE ONLY (E)

No strength training was performed.
Additional 1.5 hours of cycling training
compared to E+S was performed

% CHANGES PRE-POST THE 12 WEEKS



12 weeks of heavy gym strength training during the preparatory period is beneficial for durability in well-trained cyclists

Reference: Ronnestad et al. 2011



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3. UNILATERAL GYM STRENGTH TRAINING EXERCISES ARE BETTER THAN BILATERAL ONES WHEN TRYING TO IMPROVE CYCLING PERFORMANCE

[Research Summary and Comment](#)

UNILATERAL OR BILATERAL GYM STRENGTH EXERCISES TO IMPROVE CYCLING PERFORMANCE?

24 trained amateur triathletes and cyclists ($\text{VO}_{2\text{max}} \sim 60$) were divided into 3 different groups which trained similarly on the bike (6hrs per week) for 10 weeks with the only following difference:

UNILATERAL GYM (UG)

4x4-10 RM alternating left and right leg of the following exercises: leg-press, leg extension, and leg curl.

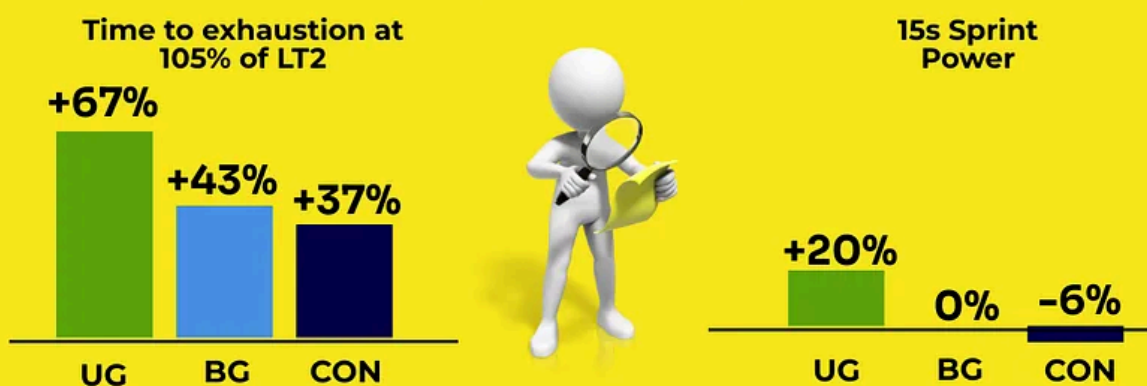
BILATERAL GYM (BG)

4x4-10 RM of the following two-legged exercises: leg-press, leg extension, and leg curl.

CONTROL (CON)

no gym strength training was performed.

% CHANGES PRE-POST THE 10 WEEKS



Unilateral gym strength training seems to be more effective than bilateral strength training to improve cycling performance.

Reference: Ji et al 2022



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4. PRACTICAL GYM STRENGTH TRAINING GUIDELINES FOR ENDURANCE CYCLISTS

[Research Summary and Comment](#)

OPTIMIZE STRENGTH TRAINING FOR CYCLING PERFORMANCE

PREPARATORY PERIOD



- » Begin immediately just at the end of the competitive season in order to not start with concurrent high load/intensity of endurance training
- » At the beginning it is common to have heavy and sore legs the first days after a strength sessions
- » First develop a proper lifting technique with lighter loads
- » Involve similar muscle groups and imitate sport specific movements
- » Focus on performing the concentric phase of the lift as quick as possible
- » 4-10 RM for 2-3 sets with 2-3' recovery
- » Daily undulated periodized program
- » 2 sessions x week for 12 weeks in the prep. period are enough to achieve a sufficient increase in performance

COMPETITIVE PERIOD



- » 1 session x week with low volume (1-2 sets per exercise) with high intensity (4-6 RM) is sufficient to maintain previous obtained strength training adaptations

Rønnestad and Mujika 2015



LITERATURE BRIEFING:

- In professional (continental level) cyclists, 2x heavy gym strength training sessions for 10 weeks during the preparatory period improve W/kg performance in efforts lasting from 1s to 40min. So, strength training can be beneficial for all rider types: sprinters, puncheurs, time trialists and climbers. (1)
- 12 weeks of heavy gym strength training during the preparatory period improve durability in well-trained amateur cyclists. (2)
- Unilateral gym strength exercises seems to be more effective than bilateral ones to improve cycling performance. (3)
- Based on a bunch of studies, practical recommendations for implementing gym strength training to optimize cycling performance can be found [here](#). (4)

REFERENCES:

1. Rønnestad BR, Hansen J, Nygaard H. 10 weeks of heavy strength training improves performance-related measurements in elite cyclists. *J Sports Sci.* 2017 Jul;35(14):1435-1441.
2. Rønnestad BR, Hansen EA, Raastad T. Strength training improves 5-min all-out performance following 185 min of cycling. *Scand J Med Sci Sports.* 2011 Apr;21(2):250-9.
3. Ji S, Donath L, Wahl P. Effects of Alternating Unilateral vs. Bilateral Resistance Training on Sprint and Endurance Cycling Performance in Trained Endurance Athletes: A 3-Armed, Randomized, Controlled, Pilot Trial. *J Strength Cond Res.* 2022 Dec 1;36(12):3280-3289.
4. Rønnestad BR, Mujika I. Optimizing strength training for running and cycling endurance performance: A review. *Scand J Med Sci Sports.* 2014 Aug;24(4):603-12.